

Ruthenium-106 plaque brachytherapy for thick posterior uveal melanomas

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ABSTRACT

Aims: Ruthenium-106 brachytherapy is an effective method for treating small to medium uveal melanomas. The purpose of this study was to examine its effectiveness and safety in the management of thick posterior uveal melanoma (apical height ≥ 8.0 mm) and to compare it with enucleation.

Methods: 126 consecutive patients with thick uveal melanoma were included. 63 patients treated with Ruthenium-106 brachytherapy were compared with 63 patients treated with enucleation. The main outcome measures were visual acuity, eye retention, local recurrence, metastases, all-cause mortality and melanoma-related mortality.

Results: Patients treated with brachytherapy were significantly younger and had significantly smaller tumours, compared with patients treated with enucleation. Tumour thickness in the brachytherapy group was 9.3 (SD 0.9) mm compared with 12.2 (1.9) mm in the enucleation group. The 5- and 10-year melanoma-related mortality was 20.5% and 46.2% for brachytherapy patients and 28.1% and 44.0% for enucleation patients ($p = 0.6$ and $p = 0.9$). When comparing 15 brachytherapy with 15 matched enucleation patients, the 5-year melanoma-related mortalities were similar, 28.6% and 33.3% respectively ($p = 0.7$). Complications associated with brachytherapy included tumour regrowth ($n = 15$), scleral melt ($n = 3$), neovascular glaucoma ($n = 5$) and vitreous haemorrhage ($n = 3$). In the brachytherapy group, no significant difference in survival was noted between patients who did and did not develop local recurrence ($p = 0.9$). Of the eyes that were initially treated with brachytherapy, 71.4% were saved from enucleation. Of these, 70.8% had a final visual acuity of 20/200 or better.

Conclusions: Ruthenium-106 brachytherapy is an alternative to enucleation in some thick posterior uveal melanomas.

Plaque radiotherapy (brachytherapy) has assumed a major role in the management of posterior uveal melanoma,¹ since it is considered an equivalent globe-sparing alternative to enucleation.² In most centres, brachytherapy is most commonly used for tumours less than 8 mm in height, while thicker tumours are conservatively managed by enucleation.¹ The maximal tumour height accepted by the Collaborative Ocular Melanoma Study (COMS) for brachytherapy with iodine-125 was 10 mm³. As for thicker tumours, concern exists in choosing brachytherapy over enucleation, for fear that sufficient radiation might not reach the tumour apex.

Several studies have reported the use of iodine-125 brachytherapy for treating thick uveal

melanomas,^{1 4-6} but there is no such report with Ru-106 brachytherapy. Ru-106 brachytherapy, first described by Lommatzsch and Vollmar,⁷ has a threefold faster dose fall-off as compared with iodine-125.⁸ The amount of radiation that can be delivered to the apex is limited by a maximum of 1000 Gy to the base, in order to minimise complications (such as scleral necrosis). Ru-106 plaque with 1000 Gy to the base conveys less than 100 Gy to 7 mm from the base,⁸ an amount generally considered to be insufficient for treatment. Insufficient radiation to the apex delivered by Ru-106 could jeopardise local tumour control and increase the risk of recurrence, as well as metastases and mortality.^{1 9-12}

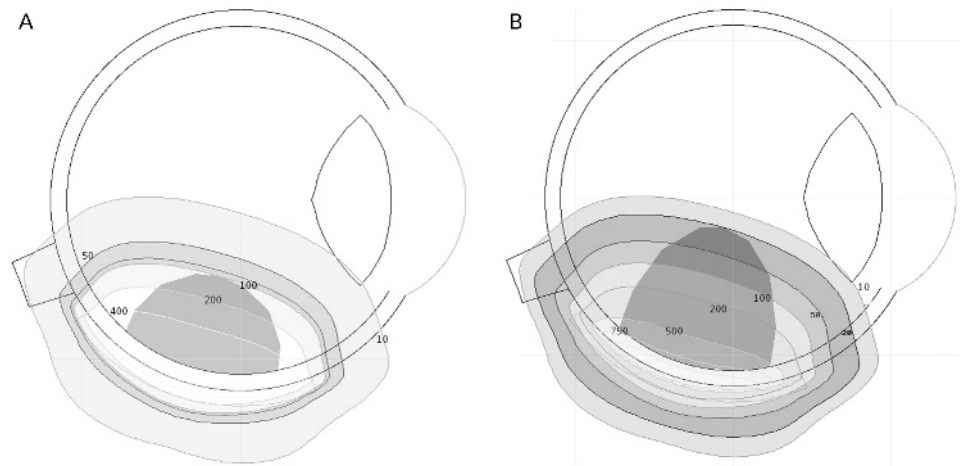
In this study, we present our experience with Ru-106 plaque brachytherapy in the management of 63 thick (more than 8 mm height) posterior uveal melanomas, comparing the results with those of enucleated eyes with tumours of similar thickness.

PATIENTS AND METHODS

One hundred and twenty-six consecutive patients (126 eyes) with thick posterior uveal melanomas (apical thickness 8 mm or more) who presented to the Ocular Oncology service at the Hadassah-Hebrew University Medical Center, between 1989 and 2007 were included in the study. Sixty-three patients were treated by enucleation, and 63 patients were treated by Ru-106 plaque brachytherapy (IBT-BEBIG, Berlin). The equal number of patients was coincidental. We included only patients who did not receive any previous treatment. The treatment was intended to deliver 100 Gy to the apex, not exceeding 1000 Gy to the base. However, Ru-106 plaque with 1000 Gy to the base conveys less than 100 Gy to 7 mm from the base, an amount generally considered to be insufficient for treatment (fig 1). The patients were informed of this limitation. We generally recommended that those patients undergo enucleation, but some patients preferred brachytherapy in an effort to preserve their eyes. Ru-106 brachytherapy was administered as an "off label" treatment and done according to the Declaration of Helsinki. Institutional Review Board approval was not required for this retrospective study (as determined by the IRB of Hadassah Medical Center).

Patients treated with brachytherapy were followed by dilated ocular exams and ocular echography (standardised A and B-scans; I³ System, Innovative Imaging, Sacramento, California¹³) 3 months after brachytherapy and every 6 months thereafter. The echographic examination included measurement of the tumours' height (A scan), base

Figure 1 Theoretical isodose curves of a Ru-106 brachytherapy plaque (CCB plaque, IBT-BEBIG) on a 6 mm tumour (A) in comparison with a tumour of 9 mm initial height (B). Both tumours received the maximal dose of 1000 Gy to the base. While the 6 mm tumours receives 100 Gy to the apex, the 9 mm tumours would receive only 20 Gy to the apex, an amount generally considered to be insufficient for treatment. Numbers on curves are in Gy. Plots were produced with the "Plaque Simulator" (IBT-BEBIG).



diameter (B scan), internal reflectivity (A scan), regularity (A scan), vascularity (A scan) and extrascleral extension (A and B scans). All patients were followed by liver-function tests and liver ultrasonography every 6 months. Patients treated with brachytherapy who suffered either local recurrence of the tumour or complications were subsequently enucleated. To examine the effect of brachytherapy on vision, we summarised the best-corrected visual acuity at the last available follow-up. Survival was ascertained by checking the national database of registered deaths. We compared melanoma-related mortality and all-cause mortality, as well as the rate of metastases.

Matched case-control substudy

Since the brachytherapy patients were significantly different from the enucleation patients in terms of age and tumour size, we selected 15 patients from each group who could be matched for age and tumour thickness. We compared final best-corrected visual acuity and survival rates in those patients.

Statistical analysis

The Student *t* test was used for continuous variables and χ^2 test with Yates correction for proportions (SPSS ver. 12 (SPSS, Chicago, Illinois)). Probabilities less than 5% were considered statistically significant. We used the product-limit Kaplan–Meier method for survival¹⁴ and compared survival rates for each prognostic parameter using the logrank test.¹⁵ We used the multivariate proportional hazards model of Cox¹⁶ to adjust for the simultaneous effects on survival of the various covariates.

RESULTS

Table 1 compares patients treated with Ru-106 brachytherapy and enucleation.

The two groups did not differ significantly in gender, laterality of the tumour or duration of follow-up. Patients treated with brachytherapy were significantly younger ($p = 0.02$) and had significantly smaller tumours ($p < 0.001$ for tumour thickness, $p = 0.0002$ for largest basal diameter) compared with patients treated by enucleation. No difference in either 5-year or 10-year mortality or metastases rates was found between the brachytherapy and enucleation groups.

Figure 2 shows the Kaplan–Meier survival curves for all-cause mortality (fig 2A) and melanoma-related mortality (fig 2B) of patients primarily treated with brachytherapy, compared with patients primarily treated with enucleation. The difference

between the curves was not statistically significant ($p = 0.84$ and $p = 0.48$, respectively, logrank test). In addition, no statistically significant difference existed for metastases rate ($p = 0.84$, logrank test). Cox proportional hazards model found no effect of the type of treatment (brachytherapy or enucleation) on survival ($p = 0.92$) after correcting for age ($p = 0.04$), largest base diameter ($p = 0.03$), tumour thickness ($p = 0.91$) and gender ($p = 0.81$).

Since the groups differed in tumour size and age, we also compared a subgroup of 15 brachytherapy and 15 enucleation patients who were matched for age and tumour thickness (table 2).

As can be seen in table 2, no difference existed in age, gender, laterality of the tumour, tumour thickness, largest basal diameter or duration of follow-up. Again, no difference in either 5-year or 10-year mortality or metastases rates was found between the matched groups. When applying Kaplan–Meier survival analysis to the subgroup of matched tumours, no significant difference was present for metastases ($p = 0.87$), all-cause mortality ($p = 0.85$), or melanoma-specific mortality ($p = 0.85$; fig 3). Similarly, the Cox proportional hazards model found no effect of the type of treatment on survival ($p = 0.95$) after correcting for age, largest base diameter, tumour thickness and gender.

On average, tumours treated with brachytherapy received 69.9 (SD 22.0) Gy (median 66.75) to the apex and 989.64 (51.3) Gy to the base (median 1000). Fifteen of the 63 tumours (23.8%) treated with brachytherapy recurred locally, 13 (20.6%) of them in the first 5 years after the treatment. The average time from brachytherapy to tumour regrowth was 38.5 (26.4) months (mean (SD)). The tumours that recurred did not differ from the tumours that did not recur as far as patients' age and gender, laterality of the tumour, initial apical height and largest basal diameter were concerned (table 3).

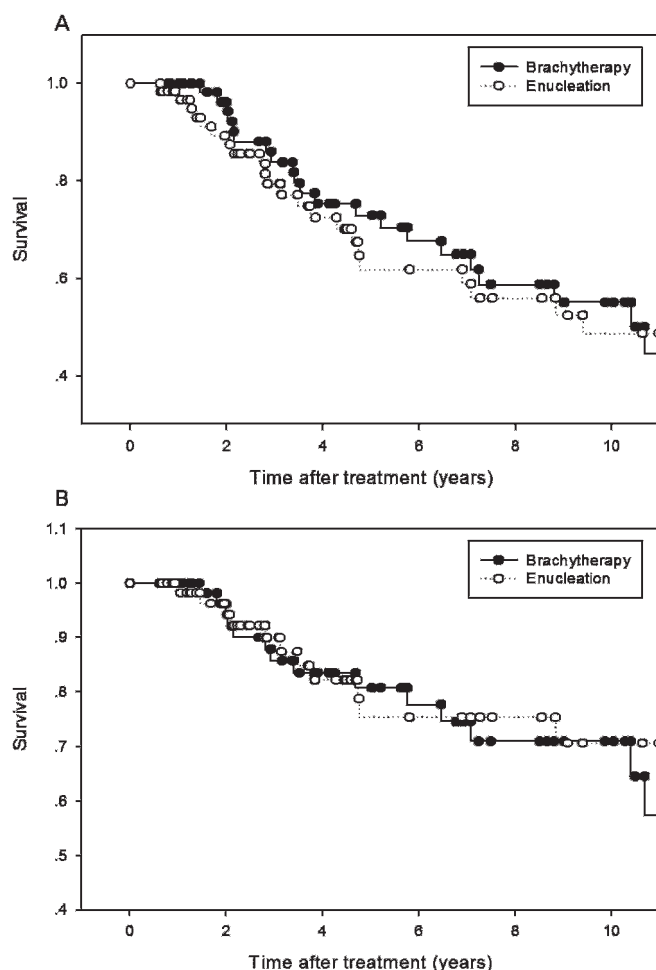
Regarding therapy, there was no difference between the tumours in the amount of irradiation to the base, but the tumours that recurred received significantly less irradiation to the apex ($p = 0.0003$). Kaplan–Meier survival analysis for all-cause mortality and melanoma-related mortality (fig 4) showed no significant difference in survival between patients who did and did not develop local recurrence ($p = 0.9$ and 0.8 , respectively, logrank test).

Eyes with tumour recurrence were either retreated with brachytherapy ($n = 5$) or enucleated ($n = 10$). Of those, three eyes had extrascleral protrusion (with scleral melting), and two eyes developed neovascular glaucoma and vitreous haemorrhage. In addition, three eyes, previously treated with brachytherapy without recurrence, were enucleated due to

Table 1 Characteristics and outcomes of all patients with thick uveal melanoma treated with either brachytherapy or enucleation

	Brachytherapy (n = 63)	Enucleation (n = 63)	p Value
Gender (males)	29 (46%)	34 (54%)	0.47
Age (years, mean (SD))	57.52 (16.80)	64.37 (16.05)	0.02
Right eyes	27 (43%)	34 (55%)	0.24
Tumour thickness (mm by A-mode, mean (SD) (range))	9.29 (0.90) (8.00 to 11.81)	12.23 (1.89) (8.00 to 19.40)	<0.001
Largest basal diameter (mm by B-mode, mean (SD) (range))	13.6 (2.8) (8.9 to 19.1)	15.9 (3.8) (7.1 to 29.2)	0.0002
Follow-up duration (months, mean (SD))	69.6 (51.7)	70.8 (62.6)	0.90
Median survival (years)	11.6	10.6	
5-year metastases rate (%)	22.5	39.5	0.16
10-year metastases rate (%)	48.1	54.8	0.80
5-year melanoma-related mortality (%)	20.5	28.1	0.64
10-year melanoma-related mortality (%)	46.2	44.0	0.90
5-year all-cause mortality (%)	27.9	42.5	0.24
10-year all-cause mortality (%)	57.6	60	0.96

χ^2 test was used for proportions and Student t test for means. Survival and mortality times are considered from the diagnosis of the ocular tumour.

**Figure 2** Kaplan–Meier survival curves comparing all patients treated with Ru-106 brachytherapy or enucleation. (A) All-cause mortality ($p = 0.84$, logrank test). (B) Melanoma-related mortality ($p = 0.48$, logrank test).

complications (all had neovascular glaucoma, one had vitreous haemorrhage). As a result, of 63 eyes that received primary treatment with brachytherapy, 45 (71.4%) were salvaged.

Visual acuity at the last follow-up was available for 48 of 63 patients treated with brachytherapy (not including enucleated eyes and failure of follow-up). Their mean (SD) best-corrected visual acuity was log MAR 1.1 (1.2) (decimal equivalent 20/250, median 20/80). Thirty-four patients (70.8%) had a best-corrected visual acuity of 20/200 or better.

DISCUSSION

Although enucleation is the most common primary management of thick (≥ 8 mm) uveal melanomas, several studies have shown gamma-radiation emitting iodine-125 brachytherapy to be a possible alternative, with regard to survival and local tumour control.^{1–4,6} No previous study has reported using beta irradiation emitting Ru-106 to treat thick uveal melanomas. The problem with beta irradiation-emitting Ru-106 might be its limited depth of penetration, which might not reach the apex of thick tumours. Ru-106, as compared with iodine-125, has a threefold faster dose fall-off.⁸ The effective amount of radiation of 100 Gy or more reaches less than 7 mm when Ru-106 plaque with 1000 Gy to the base is used.⁹ Thus, tumours thicker than 7 mm might receive an insufficient dose of less than 100 Gy to their apices. On the other hand, for the same radiation dose delivered to the apex, the base irradiated by Ru-106 receives up to three times more radiation than I-125. This might help cause extensive damage to the blood vessels travelling through the base to supply the tumour.

We have shown here that thick uveal melanomas (initial apical height of 8 mm and greater) that were treated with Ru-106 brachytherapy had 5- and 10-year mortalities statistically comparable with thick uveal melanomas primarily treated with enucleation. Moreover, the 5-year and 10-year metastases rates were comparable between the two groups (although slightly higher in the enucleation group, but not statistically significant). Those results persisted when the two groups were matched for age and tumour thickness.

Table 2 Characteristics of matched thick uveal melanoma patients treated with brachytherapy and enucleation

	Brachytherapy (n = 15)	Enucleation (n = 15)	p Value
Gender (males)	5 (33.3%)	7 (46.7%)	0.70
Age (years, mean (SD))	54.4 (14.7)	56.7 (16.2)	0.68
Right eyes	6 (40%)	8 (53%)	0.72
Tumour thickness (mm by A-mode, mean (SD))	10.5 (0.6) (9.0 to 11.8)	10.4 (0.8) (8.4 to 11.3)	0.70
Largest basal diameter (mm by B-mode, mean (SD) (range))	13.8 (2.2) (9.8 to 16.6)	14.7 (4.6) (10.5 to 17.8)	0.50
Follow-up duration (months, mean (SD))	59.8 (45.6)	75.6 (65.1)	0.44
5-year metastases rate (%)	28.6	40	0.97
10-year metastases rate (%)	50	57.1	0.76
5-year melanoma-related mortality (%)	28.6	33.3	0.73
10-year melanoma-related mortality (%)	50	50	0.56

χ^2 test was used for proportions and Student t test for means. In both groups, all-cause mortality = melanoma-related mortality. Survival and mortality times are considered from the diagnosis of the ocular tumour.

Our results are comparable with the 5-year large-melanoma-specific mortality of 26%, reported in a meta-analysis of long-term survival after Ru-106 brachytherapy,¹⁷ and with the 5-year large-melanoma-specific mortality of 28% as reported for enucleation in the COMS with iodine-125.¹⁸

In our study, the primary enucleation patients were significantly older than the brachytherapy patients, perhaps because younger patients might be more reluctant to remove their eye. Primary enucleation patients also had significantly thicker tumours at presentation. For comparison, in the enucleation group the thickest tumour was 19.4 mm high, while in the brachytherapy group it was 11.8 mm high. Since the enucleation group patients were older, tumours might have had more time to grow before being detected, possibly because older patients have other ocular conditions, such as cataracts or retinal diseases, which decrease visual acuity, making visual symptoms caused by the tumour more subtle. Moreover, very thick tumours would be more readily enucleated. After correction for tumour size and patient age, the metastasis and the mortalities were not significantly different between enucleation and brachytherapy patients.

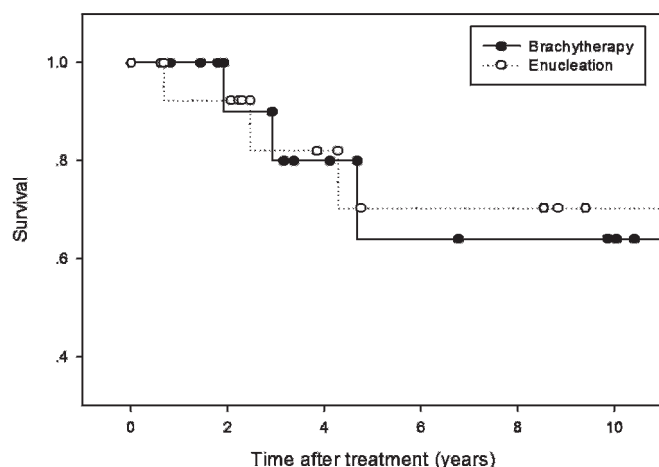


Figure 3 Kaplan-Meier survival curves comparing melanoma-related mortality in 15 matched patient pairs treated with Ru-106 brachytherapy or enucleation ($p = 0.85$, logrank test).

The 5-year local recurrence rate after primary Ru-106 brachytherapy was 23.8% in our study, compared with 6–9% local recurrence reported in the literature for thick uveal melanomas treated with iodine-125 brachytherapy.¹⁴ This is particularly worrisome in that failure of local control may contribute to a risk for metastasis. However, in this study we did not observe an increased metastasis rate in those patients who had local recurrence. The higher recurrence rate could be a result of the limited penetration of the β irradiation. Indeed, we found a significantly lower irradiation dose reaching the apex of those tumours that did recur (table 3). The tumours that did recur had a longer follow-up. This is very much expected, as a longer follow-up would result in more recurrence. Despite the longer follow-up of patients with local recurrence, no significant difference in survival was found between patients who did and did not experience local tumour recurrence after brachytherapy (5-year and 10-year survival as well as Kaplan-Meier survival analysis). The similarity in survival could be attributed to the similar (and high) irradiation dose given to the base (table 3), which might have obliterated blood supply to the tumour and prevented further spread of the disease.

In this study, in spite of some complications, such as local recurrence, scleral necrosis and neovascular glaucoma, 71.4% of eyes initially treated with brachytherapy could be salvaged from enucleation. This result is comparable, with 76–90.3% eye retention reported after iodine-125 brachytherapy for thick melanomas,¹⁴ and is predictably lower than the 82–87.5% reported after Ru-106 brachytherapy for any-thickness melanomas.^{3 17 19}

In the literature, the best-corrected visual acuity of 20/200 or worse following iodine-125 brachytherapy ranges between 57% and 94%.^{1 5 9} In this study only 18 of 48 (37.5%) patients treated with Ru-106 brachytherapy had best-corrected visual acuity of 20/200 or worse at their last follow-up. This might imply that the limited penetration of the Ru-106 irradiation might be more beneficial for final visual acuity. This assumption is supported by Wilkinson *et al*²⁰ who have shown that the mean dose to the macula, disc and lens was 18%, 53% and 89% lower, respectively, with Ru-106 than with I-125.

Although Ru-106 brachytherapy might not reach the apex of the tumour with a sufficient irradiation dose, its effectiveness on thick melanomas could be explained by vascular damage to the tumour base due to the high amount of radiation delivered to the base. Radiation-induced thrombosis and necrosis of small blood vessels at the base would cause ischaemia in the apex.

Table 3 Comparison of tumours treated with brachytherapy that did and did not recur locally

	Local recurrence (n = 15)	No local recurrence (n = 48)	p Value
Gender (males)	6 (40%)	23 (47.9%)	0.8
Age (years, mean (SD))	52.26 (14.05)	59.03 (16.30)	0.15
Right eyes	7 (46.7%)	20 (41.7%)	0.9
Tumour thickness (mm by A-mode, mean (SD))	9.07 (0.63) (8.1 to 10.3)	9.35 (0.97) (8.0 to 11.8)	0.3
Largest basal diameter (mm by B-mode, mean (SD) (range))	13.8 (4.1) (8.9 to 19.1)	13.5 (2.2) (9.5 to 17.2)	0.71
Radiation to base (Gy)	971.1 (50.5)	997.0 (50.4)	0.08
Radiation to apex (Gy)	52.8 (4.9)	75.6 (22.7)	0.0003
Follow-up duration (months, mean (SD))	92.4 (48.9)	48.7 (54.2)	0.0016

χ^2 test was used for proportions and Student t test for means.

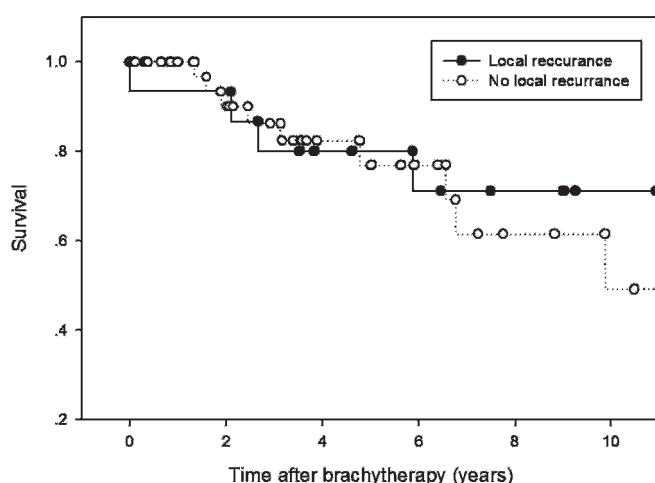


Figure 4 Kaplan–Meier survival curves comparing melanoma-related mortality among patients who had or did not have local recurrence after initial treatment with Ru-106 brachytherapy ($p = 0.8$, logrank test).

Pathology slides of enucleated eyes which were previously treated with brachytherapy show extensive vascular damage.^{21 22}

In conclusion, our results indicate that Ru-106 brachytherapy is an alternative for treating selected thick posterior uveal melanomas. A higher-than-usual recurrence rate, which does not compromise survival, should be taken into consideration.

Competing interests: None.

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